Cosets, Lagrange's Theorem, Normal Subgroups and Quotient Groups

From the class book: Chapter 8, Sections 8.1 to 8.4 and Chapter 9, Sections 9.1 to 9.3

1. Key definitions

Cosets and index. For $H \leq G$ and $a \in G$, the left coset is $aH = \{ah : h \in H\}$ and the right coset is $Ha = \{ha : h \in H\}$. The number of distinct left cosets (always equal to the number of right cosets) is the index $[G : H]$.

Group action. A map $G \times X \to X$, $(g, x) \mapsto g \cdot x$, with $e \cdot x = x$ and $g \cdot (h \cdot x) = (gh) \cdot x$. The orbit of x is $\mathrm{Orb}(x) = \{g \cdot x : g \in G\}$; the stabilizer $\mathrm{Stab}(x) = \{g : g \cdot x = x\}$ is a subgroup of G.

Normal subgroup. $N \leq G$ is normal, $N \trianglelefteq G$, if $gNg^{-1} = N$ for every $g \in G$.

Factor (quotient) group. For $N \trianglelefteq G$, the set G/N of cosets with $(aN)(bN) = abN$; identity N, inverse $(aN)^{-1} = a^{-1}N$.

Commutator subgroup. G' is the subgroup generated by all commutators $[a, b] = aba^{-1}b^{-1}$.

Simple group. A group $G \neq \{e\}$ whose only normal subgroups are $\{e\}$ and G.

2. Main results

Theorem (Properties of cosets)
Let $H \leq G$, $a, b \in G$.

- $a \in aH$; $aH = H \iff a \in H$; aH is a subgroup $\iff a \in H$.
- $aH = bH \iff a^{-1}b \in H$; $Ha = Hb \iff ab^{-1} \in H$.
- Two left cosets are equal or disjoint, so the left cosets partition G; all have $|H|$ elements.
- $aH = Ha \iff aHa^{-1} = H$. The map $aH \mapsto Ha^{-1}$ is a bijection from left to right cosets.

Theorem (Lagrange)
If G is finite and $H \leq G$, then $|H|$ divides $|G|$ and $[G : H] = |G|/|H|$.
Idea: the $[G : H]$ left cosets partition G and each has $|H|$ elements.

Consequences. (i) $o(a) \mid |G|$, so $a^{|G|} = e$. (ii) A group of prime order is cyclic. (iii) Euler: $a^{\varphi(n)} \equiv 1 \pmod{n}$ when $\gcd(a, n) = 1$ (apply (i) to $U(n)$); Fermat is $n = p$. (iv) Tower law: $K \leq H \leq G \Rightarrow [G : K] = [G : H][H : K]$. (v) If $\gcd(|H|, |K|) = 1$ then $H \cap K = \{e\}$.

Theorem (Orbit–Stabilizer)
For a finite group acting on X: $|\mathrm{Orb}(x)| = [G : \mathrm{Stab}(x)]$, i.e. $|G| = |\mathrm{Orb}(x)| \cdot |\mathrm{Stab}(x)|$. The orbits partition X. (Burnside: number of orbits = average number of fixed points.)

Theorem (Tests for normality)
For $N \leq G$ the following are equivalent: $N \trianglelefteq G$; $gng^{-1} \in N$ for all $g \in G, n \in N$; $gN = Ng$ for all g; $(aN)(bN) = abN$ is well defined; N is a union of conjugacy classes; N is the kernel of some homomorphism.

Theorem (Factor groups)
If $N \trianglelefteq G$, G/N is a group and $|G/N| = |G|/|N|$ for finite G.

- G abelian $\Rightarrow$ G/N abelian; G cyclic $\Rightarrow$ G/N cyclic (converses fail: $S_n/A_n \cong \mathbb{Z}_2$).
- $o(aN)$ divides $o(a)$; if $[G:N] = m$ then $x^m \in N$ for all x.
- If $G/Z(G)$ is cyclic then G is abelian. Hence for non-abelian G, $[G:Z(G)]$ is never prime.

Theorem (Commutator subgroup)
$G' \trianglelefteq G$ and, for $N \trianglelefteq G$, G/N is abelian $\iff$ $G' \subseteq N$. So G/G' is the largest abelian quotient, and G is abelian $\iff$ $G' = \{e\}$.

Theorem (Simple groups)
An abelian group is simple $\iff$ it is $\mathbb{Z}_p$, p prime. A_n is simple for $n \geq 5$; A_5 (order 60) is the smallest non-abelian simple group. S_n ($n \geq 3$) and A_4 are not simple. N is a maximal normal subgroup $\iff$ G/N is simple.

Theorem (Small orders)
A group of order p^2 is abelian ($\mathbb{Z}_{p^2}$ or $\mathbb{Z}_p \times \mathbb{Z}_p$). A group of order $2p$ (p odd prime) is $\mathbb{Z}_{2p}$ or D_p, and has a unique (hence normal) subgroup of order p. A group of order pq, $p < q$, $p \nmid q - 1$, is cyclic.

3. Standard tables

Number of groups of order n (up to isomorphism)

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
all	1	1	1	2	1	2	1	5	2	2	1	5	1	2	1	14
abelian	1	1	1	2	1	1	1	3	2	1	1	2	1	1	1	5

Order 4: $\mathbb{Z}_4, K_4$. Order 6: $\mathbb{Z}_6, S_3$. Order 8: $\mathbb{Z}_8,\ \mathbb{Z}_4 \times \mathbb{Z}_2,\ \mathbb{Z}_2^3,\ D_4,\ Q_8$. Order 12: $\mathbb{Z}_{12},\ \mathbb{Z}_6 \times \mathbb{Z}_2,\ A_4,\ D_6$, and the dicyclic group $\mathbb{Z}_3 \rtimes \mathbb{Z}_4$.

Subgroups and normal subgroups

Group	Subgroups	Normal	Group	Subgroups	Normal
S_3	6	3	Q_8	6	6
D_4	10	6	D_5	8	3
D_6	16	7	A_4	10	3
S_4	30	4	$S_n,\ n \geq 5$	–	3

Normal subgroups: of S_n ($n \neq 4$): $\{e\}, A_n, S_n$; of S_4: $\{e\}, V_4, A_4, S_4$; of A_4: $\{e\}, V_4, A_4$; of A_n ($n \geq 5$): $\{e\}, A_n$. Here $V_4 = \{e, (12)(34), (13)(24), (14)(23)\}$.

Commutator subgroups and standard quotients

G	S_3	S_4	A_4	Q_8	D_4	S_n ($n \geq 2$)
G'	A_3	A_4	V_4	$\{\pm 1\}$	$\{e, r^2\}$	A_n
G/G'	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_3$	K_4	K_4	$\mathbb{Z}_2$

$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$; $\mathbb{Z}_{12}/\langle\bar{4}\rangle \cong \mathbb{Z}_4$; $GL(n,\mathbb{R})/SL(n,\mathbb{R}) \cong \mathbb{R}^*$; $\mathbb{R}/\mathbb{Z} \cong S^1 = \{z : |z| = 1\} \cong \mathbb{C}^*/\mathbb{R}^+$; $Q_8/\{\pm 1\} \cong K_4$; $S_4/V_4 \cong S_3$; $A_4/V_4 \cong \mathbb{Z}_3$.

Two special abelian groups

- $\mathbb{Q}/\mathbb{Z}$: infinite, every element of finite order, $o(\frac{m}{n}+\mathbb{Z}) = n$ when $\gcd(m,n)=1$; exactly $\varphi(n)$ elements of order n and exactly one subgroup of order n (cyclic, generated by $\frac{1}{n}+\mathbb{Z}$). Not finitely generated; every proper subgroup has infinite index. The Prüfer 2-group $\{\frac{m}{2^k}+\mathbb{Z}\}$ is infinite but all its proper subgroups are finite cyclic.
- $(\mathcal{P}(\mathbb{N}), \triangle)$: every element has order ≤ 2, so abelian; finite subgroups have order 2^k.

4. Exam shortcuts and standard counter-examples

Exam shortcut. Normal for free: subgroups of index 2 (then $x^2 \in H$ for all x), the centre, G', kernels, every subgroup of an abelian group, and a subgroup that is the only one of its order. Also, if $x^2 \in H$ for all $x \in G$, then $H \trianglelefteq G$.

Converse of Lagrange fails. A_4 (order 12) has no subgroup of order 6. The converse does hold for cyclic groups, finite abelian groups and finite p-groups; Sylow gives every prime-power order.

Normality is not transitive. $\langle (12)(34)\rangle \trianglelefteq V_4 \trianglelefteq A_4$, but $\langle (12)(34)\rangle$ is not normal in A_4. Also $\{e, (12)\}$ is not normal in S_3: $(13)(12)(13) = (23)$.

All subgroups normal does not mean abelian. Q_8 is non-abelian, yet all 6 of its subgroups are normal (Hamiltonian group). Isomorphic subgroups need not be conjugate: the three subgroups of order 2 in K_4.

Quick tests. "Which cannot be the order of a subgroup/element?" – pick a non-divisor of $|G|$. $a^{|G|} = e$ always, but $|G|$ need not be the order of any element. $G/Z(G)$ can never be a non-trivial cyclic group; for non-abelian groups of order 8, $G/Z(G) \cong K_4$. Class equation of A_5: $60 = 1 + 15 + 20 + 12 + 12$; no union of classes containing 1 has size dividing 60 except 1 and 60, so A_5 is simple.

5. Solved examples

Example 1. Find the remainder when 2^{2026} is divided by 15.

Solution. $U(15)$ has order $\varphi(15) = 8$, so $2^8 \equiv 1$. In fact $2^4 = 16 \equiv 1 \pmod{15}$. Since $2026 = 4 \cdot 506 + 2$, $2^{2026} \equiv 2^2 = 4$. **Answer: 4.**

Example 2. In S_3 let $H = \{e, (12)\}$. List the left and right cosets of H and decide whether $H \trianglelefteq S_3$. (Compose right to left.)

Solution. $[S_3 : H] = 3$. Left cosets: H, $(13)H = \{(13), (13)(12)\} = \{(13), (123)\}$, $(23)H = \{(23), (132)\}$. Right cosets: H, $H(13) = \{(13), (12)(13)\} = \{(13), (132)\}$, $H(23) = \{(23), (123)\}$. Since $(13)H \neq H(13)$, H is not normal. (Index 3, so the "index 2" shortcut does not apply.)

Example 3. Let $H = \langle (1,2)\rangle \leq \mathbb{Z}_4 \times \mathbb{Z}_4$. Find the order of G/H and identify it.

Solution. $H = \{(0,0), (1,2), (2,0), (3,2)\}$, so $|G/H| = 16/4 = 4$. The coset $(0,1) + H$: $2(0,1) = (0,2) \notin H$, so its order is 4. Hence G/H is cyclic: $G/H \cong \mathbb{Z}_4$ (not K_4).

Example 4. Let G be a non-abelian group of order p^3. Find $|Z(G)|$ and $G/Z(G)$.

Solution. The centre of a non-trivial p-group is non-trivial, so $|Z(G)| \in \{p, p^2, p^3\}$. Not p^3 since G is non-abelian. If $|Z(G)| = p^2$, then $|G/Z(G)| = p$, so $G/Z(G)$ is cyclic and G would be abelian – contradiction. Hence $|Z(G)| = p$ and $G/Z(G)$ has order p^2 and is not cyclic, so $G/Z(G) \cong \mathbb{Z}_p \times \mathbb{Z}_p$. (For $p = 2$: $D_4/Z(D_4) \cong Q_8/Z(Q_8) \cong K_4$.)

Example 5. S_4 acts on its subgroups by conjugation. How many conjugates does $H = \langle (1234) \rangle$ have? Also find the order of the rotation group of a regular tetrahedron.

Solution. The stabilizer of H is its normalizer $N(H)$. $N(H)$ contains the dihedral group $D_4 = \langle (1234), (13) \rangle$ of order 8, and $N(H) \neq S_4$ (since $(12)H(12) = \langle (2134) \rangle \neq H$); as $8 \mid |N(H)| \mid 24$, $|N(H)| = 8$. Orbit size $= 24/8 = 3$: the three cyclic subgroups of order 4. Tetrahedron: rotations move any vertex to any of the 4 vertices (orbit 4); the stabilizer of a vertex is the 3 rotations about its axis. Order $= 4 \cdot 3 = 12$ (it is A_4).

Example 6. Find all $N \trianglelefteq S_4$ such that S_4/N is abelian. Is S_4/V_4 abelian?

Solution. S_4/N is abelian $\iff N \supseteq S_4' = A_4$. The normal subgroups of S_4 are $\{e\}, V_4, A_4, S_4$; those containing A_4 are A_4 and S_4 (quotients $\mathbb{Z}_2$ and trivial). Since $V_4 \not\supseteq A_4$, S_4/V_4 is non-abelian; indeed $S_4/V_4 \cong S_3$. **Answer: $N = A_4$ or S_4.**